

Scalability & Future Plan

Date	12 July 2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High/Medium/Low)
1	CSV-based storage doesn't scale well beyond a single-machine prototype.	Concurrent writes from many students could corrupt or slow down the log file.	Medium
2	The Gemini API is not yet wired in; guidance is currently static per emotion via the mock handler.	Responses don't yet reflect the specific nuance of each student's actual text.	High
3	No user authentication or per-student history exists.	Can't track an individual student's progress across multiple sessions.	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single local Streamlit session, informally tested	Deploy to Streamlit Community Cloud or a small cloud VM with session isolation for classroom-scale concurrent use
2	Data Storage	Flat CSV file (data/interactions_log.csv)	Migrate to a lightweight database (e.g., SQLite, then PostgreSQL) as interaction volume grows
3	Performance	Both BiLSTM and BERT models loaded fully at Streamlit startup	Cache models with st.cache_resource and consider a lighter model for real-time inference
4	Security	No authentication; Gemini key stored only in a local .env	Add basic student login and move secrets to a proper secrets manager for hosted deployment

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
-------	-------------------------------	-----------------	-----------------

Phase 1	Consolidate mock model conflicts and unify the logger schema	Aug 2026	Reliable, consistent emotion detection and clean logging data
Phase 2	Wire in the real Gemini API and replace the BERT mock with the trained model	Sep 2026	Genuinely personalized, input-specific guidance
Phase 3	Migrate storage to SQLite/PostgreSQL and add basic student login	Oct 2026	Supports per-student history and safer concurrent access
Phase 4	Deploy to a hosted environment with classroom-scale load testing	Nov 2026	Validated to support a full class using the tool concurrently